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Design and Analysis of an Under Water Visible Light MIMO Communication System with a Camera Receiver

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Abstract – **In this paper, design, implementation and analysis of a new underwater multiple input multiple output visible light communication (VLC) system is presented. The VLC system consists of a transmitter, water channel, and a receiver. The transmitter is made out of light emitting diodes (LEDs'), while a camera based architecture is adopted at the receiver. We present a possible system design and characterize the channel for symbol detection. At the training phase, the underwater channel is characterized using a channel matrix, which presents the relationship between the multiple input LED states and the pixel values of the frame in the video stream obtained from the camera receiver. Thereafter the inverse model of the system is obtained. Then symbol detection is carried out by analysing consecutive frames of the real-time video stream. Also we analyse the performance of this system in terms of the bit error rate (BER), bit rate, behaviour of the system at different transmission distances, and present results using the experimental setup. From our study, it was realised that, reasonable BER performance can be achieved using the proposed system designs at low and medium transmission distances compared to most of the existing underwater communication systems.**

Keywords— **Visible light communications, underwater communications, multiple input multiple output, Light emitting diode, Camera receiver.**

I. Introduction

Visible light communication (VLC) systems have attracted lot of research in the near past due to its wide range of applicability and its high bandwidth performance at significantly low cost. Three possible methods exists for underwater wireless communication. They are radio frequency (RF) communication, acoustic communication and visible light communication. RF communication means utilizes electromagnetic waves in the frequency range from 3 kHz to 300 GHz for wireless transmission. RF waves are widely used in overwater applications. However applicability of RF waves in underwater communication is restricted due to high attenuation of RF waves in water [1]. Tremendous progress has been made in the field of underwater acoustic communication. There are several ways of employing such communication, but the most common is by using hydrophones. Hydrophone is a microphone which is designed for recording or listening of underwater sound. However, acoustics performance is limited by low bandwidth, high transmission losses, time varying multi-path propagation and high latency [2].

Visible light is a portion in the electromagnetic spectrum which the human eye is capable of seeing. They have wave length in the region of 400nm to 700nm. Light pulses propagating in aquatic medium suffer from attenuation and absorption. But due to the dual nature of light (i.e., ability to act as wave and as a collection of photons), light propagation in underwater is subject to less attenuation, compared to RF waves. Furthermore, the data speed achievable using VLC techniques are much higher than the data speeds achievable using acoustic transmission techniques in the water. Therefore, VLC is a possible way to realize a future high-speed underwater communication network.

A traditional underwater VLC system will consist light emitting diodes (LED's) at the transmitter end and photo diodes at the receiver end. Voltages detected at the receiver are amplified to decode the transmitted data. On off Keying (OOK), frequency shift keying (FSK) and orthogonal FSK are some widely used modulation techniques. Ambient light cancellation schemes, cross talk modelling for inter channel interference cancellation are studied extensively in overwater VLC systems. However still VLC is only possible to cater short and medium range distance transmission, only at light of sight scenarios [2], [3]-[5].

From RF technologies, it has become evident that multiple input multiple output (MIMO) techniques can be used to increase data rates of transmission. Some MIMO schemes are being proposed to increase data rates of traditional VLC systems as well [6], [7]. In a MIMO VLC system multiple LEDs at the transmitter and multiple photo diodes at the receiver are used and all photo detector signals at the receiver end are combined to optimize the data speed while minimizing errors. Unlike radio frequency communication, VLC is not omnidirectional. Thus it could benefit from a little inter channel interference when VLC principles are used in a MIMO system. This study is conducted in order to take advantage of this property of visible light in a MIMO underwater communication system.

In this paper we propose a new underwater visible light MIMO communication system in which transmitter is made of LEDs and the receiver is a camera [8]-[10]. Data communication happens from a submerged system (transmitter) to a floating body (receiver) using line of sight scenario [11]. Transmittable data are encoded into LED

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patterns, and those LED patterns are transmitted through the medium. At the receiver those LED pattern are captured via a camera and data are decoded after analysing the video stream [9], [12].

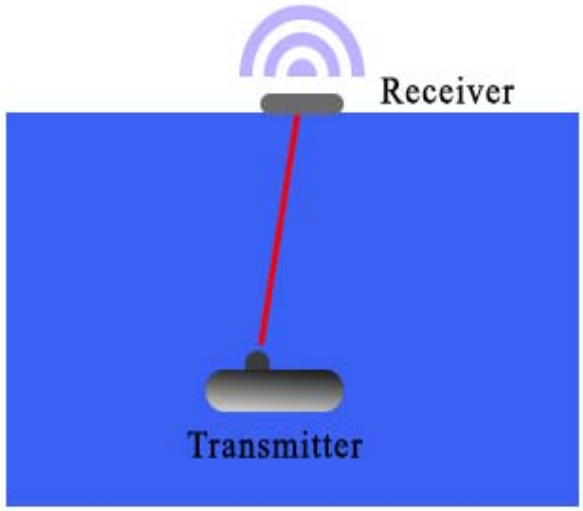

Fig. 1 Communication from the submerged body to the floating receiver using line of sight scenario.

From our study it was evident that, even though the proposed VLC system does yield comparatively less transmittable data rates, compared to a traditional over water MIMO VLC system, bit error rate performance can be optimized using a proper detection algorithm based on image processing. Therefore our design could be used for applications in which more priority is given for accuracy than the high data rates.

The remainder of the paper is organized as follows. Section 2 discuss briefly of a possible system design. Section 3 presents a channel identification technique, Section 4 discuss the way in which output detection is carried out, Section 5 evaluate the performance of the proposed system and Section 6 contains a discussion on results obtained from measurement. Finally, Section 7 gives a summary and conclusion.

II. Design of the Underwater VLC System

The VLC system is made out of the transmitter, water channel and receiver. We consider the design of the data link from a submerged body (transmitter) to a floating body (receiver) which is floating on the water surface. Transmitter and the receiver are vertically aligned and the receiver camera was focused towards the transmitter LEDs. The transmitting medium, which was still water was maintained at no turbulent condition (i.e., the water is free of any wave movement and air bubbles). The data to be transmitted was encoded into multiple data streams at the transmitter. At the receiver, the camera captures the continuous LED pattern variations in video format and the corresponding video stream is analysed by frame by frame to decode the data.

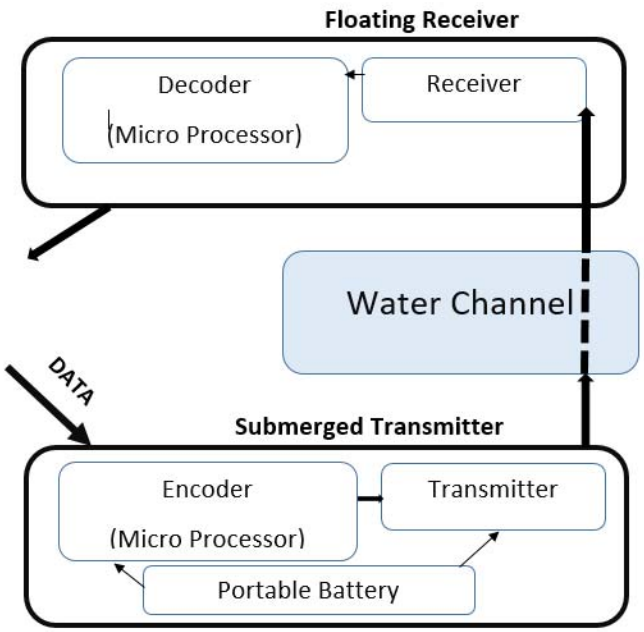

Fig. 2 Underwater VLC System with LED transmitter and camera receiver.

A. Transmitter

Transmitter of our communication system is made of LEDs. Oval-top LED, flat-top LED and power LED are used as LED types in typical VLC applications. However the spreading angle of Power LEDs was high compared to the other two LED types. Hence flat top and oval top LEDs were used in transmitter designs to reduce inter channel interferences.

Each LED of the VLC system represent separate data stream. In order to minimize the effects of cross channel interference, LEDs were designed by considering spatial and spectral separation [13], [14]. As the receiver is a camera, spatial positioning of LEDs would significantly affect the detection performance at the receiver. Thus different LED positioning setups were considered and optimized for performance. For example we considered setups where, LED placement was done at the edge of a square, along the circumference of a circle etc. However optimal spatial separation between three LEDs were achieved when they are kept at the edges of an equilateral triangle. Hence a possible transmitter LED positioning setup when more than three LEDs are used can be, any geometrical shape which is developed based on the basic triangle pattern, such as a larger triangle, parallelogram or a hexagonal.

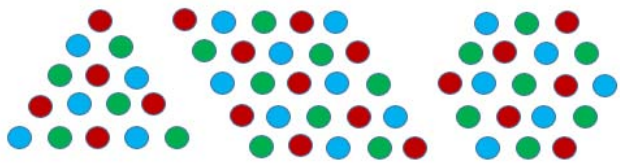

Fig. 3 Possible LED placement setups in the transmitter

Also to achieve spectral separation between adjacent LEDs, adjacent LEDs were assigned different colours. To this end, three colours which are at the three corners of the HSV

colour wheel can be used. Preferably red, green and blue are be used which facilitates colour filtering at the receiver end.

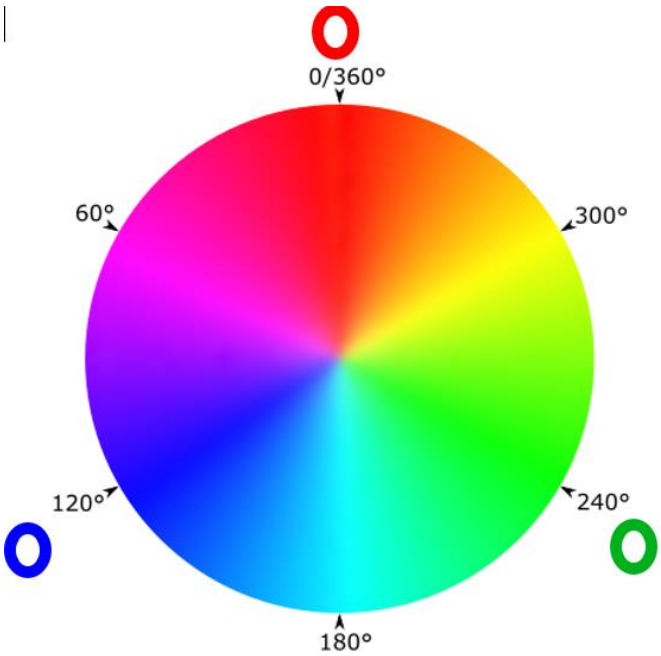

Fig 4 HSV colour wheel with three preferable colours.

As shown in figure 5(a) when transmission distance increase, occupied region by the transmitter LEDs decreases. Hence maximum size of the transmitter LED region can be calculated from (1) and (2).

Consider αx, αy to be the angle of view of the camera in x direction and y direction respectively and the minimum transmission distance is *dmin*. From simple geometry it can be shown that,

$$amax = 2 \times dmin \times \tan(\propto x) \quad (1)$$
$$bmax = 2 \times dmin \times \tan(\propto y) \quad (2)$$

In (1) and (2), *amax, bmax* are the maximum dimension of the rectangle into which all transmitter LED would fit into ($a > b$).

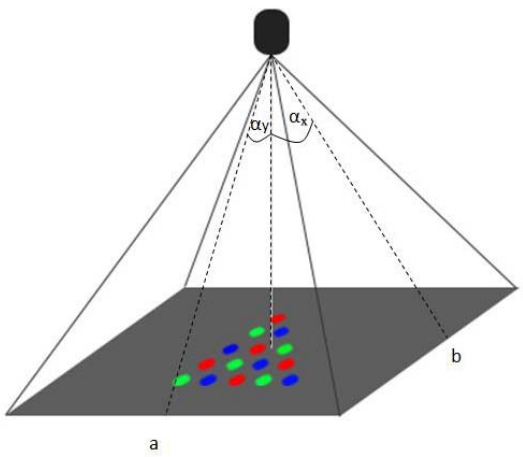

Fig. 5(a) Angle of view in 3D space.

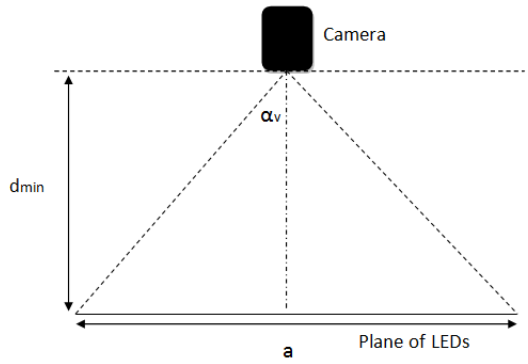

Fig. 5(b) Angle of view in 2D space

B. Receiver

A conventional camera which has high and fixed frame rate, higher angle of view and manual focus option was preferred. As the camera was focused towards the transmitter LEDs, LEDs some in ON condition and some other in OFF condition were captured by the receiver camera, and these details were stored in individual frames of the video stream. However when frames were analysed, it was evident that not all pixels in all frame were affected by the transmitter LEDs. Hence, only the region of interest in the captured frames were used for analysis, instead of the total captured frame.

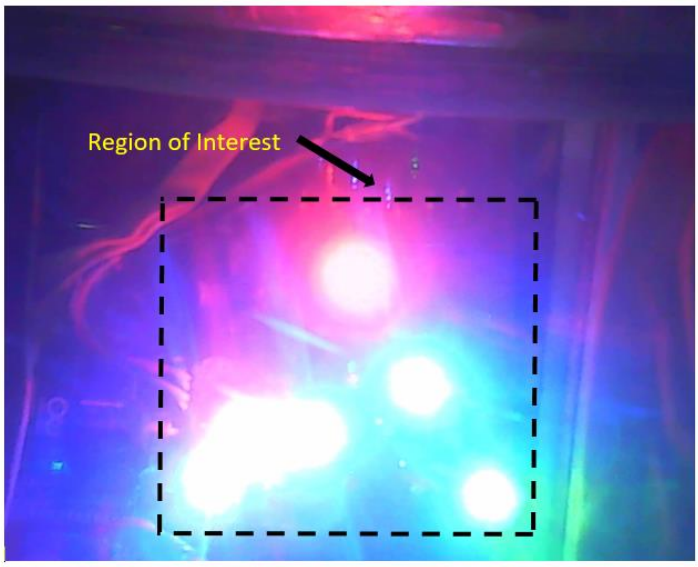

Fig. 6 A sampled captured frame and region of Interest.

C. Experimental Setup

A controlled experimental setup was used for validation. Transmitter was made of a triangle shaped LED array. The three corner LEDs are always kept ON to detect the area of interest in the captured frames. As the receiver Logitech c170 webcam was used. It had approximately 60˚ angle of view and 15fps frame rate. Maximum and minimum

transmission distances were restricted to 0.2 m and 1.0 m respectively.

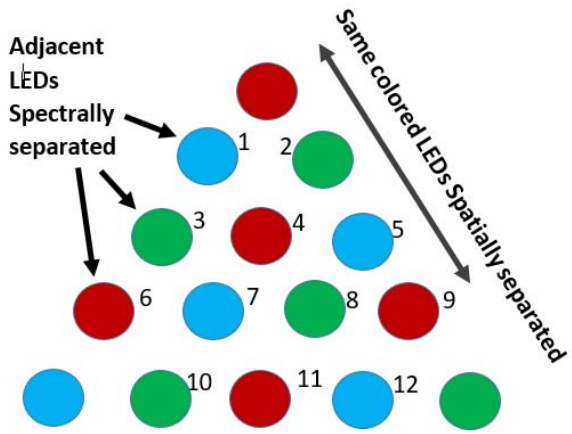

Fig.7 Experimental Setup of Transmitter with LEDs numbered.

The transmitter encodes specific number of bits to each LED pattern continuously. And as the LED pattern switching rate is equal to frame rate of the camera, each LED pattern data will be available in different frames of the video stream. So when a specific frame is analysed at the receiver, data corresponding to that frame can be decoded. Moreover, by repetitively analysing each frame individually, the total data set can be decoded. Therefore going forward, in Section 3 we identify the channel by taking each frame image as individual inputs and in Section 4, we discuss details related to output detection corresponding to a single frame.

III. Channel Identification

The input output relationship of a MIMO system can be expressed as,

$$\boldsymbol{Y} = \boldsymbol{H} * \boldsymbol{X} + \boldsymbol{N} \qquad (3)$$

where $\boldsymbol{X}$ is the input vector containing input symbols at a given instant, $\boldsymbol{Y}$ is the output vector containing output symbols, $\boldsymbol{H}$ is the channel matrix and $\boldsymbol{N}$ is the matrix which models the effect of additive noise .

In the considered VLC system, each LED data channel was considered as an input. And each pixel in the region of interest in a captured frame was considered as an output. Under no noise condition, zero water turbulence and for a fixed transmission distance the input output relationship corresponding to one single frame of the underwater VLC system was modelled using the above equation with N as a null vector.

So if a system has r LEDs, which represent separate input data channels, and if $m \times n$ is the pixel dimension of the region of interest in the camera frame, then system matrix $\boldsymbol{H}$ would be of size, $mn \times r$, where $(i,j)^{th}$ element of the channel matrix $\boldsymbol{H}$ represent the effect of jth input towards i^{th} output. Also i^{th} row of channel matrix will represent the effect of all inputs to i^{th} output. Similarly, j^{th} column of the channel matrix represent the effect towards all outputs from the jth input.

If the transmission environment does not change throughout the transmission process (same distance of transmission, no noise, zero water turbulence), channel matrix calculated would be same for any set of data transmission.

In order to proceed, first the channel matrix should be identified. For this purpose a training input data stream was transmitted and based on the outputs observed, the elements of the channel matrix were identified. The LED patterns in the training input data streams were selected so that, from every LED pattern, one column of the channel matrix can be identified. In the ith LED pattern in the training data stream, only the ith position LED and the three corner LEDs were kept ON. So the input vector corresponding to the ith LED pattern can be written as,

$$\boldsymbol{Xi} = \begin{cases} 1 & ith\ element \\ 0 & elsewhere \end{cases} \qquad (4)$$

If $\boldsymbol{Yi}$ is the output vector obtained from the pixels of the frame capturing the ith LED pattern, then i^{th} column element values of the channel matrix can be identified from the output vector using (5).

$$\boldsymbol{H}(:,\boldsymbol{i}) = \boldsymbol{Yi} - \boldsymbol{Y0} \qquad (5)$$

where $\boldsymbol{Y0}$ is the output vector obtained from the pixels of the captured frame when no input was fed into the system. So if the training data stream includes data patterns satisfying input vector of (4) for all i from one to r, from all LED patterns, all column elements can be identified as follows.

$$\boldsymbol{Xi} = \begin{cases} 1 & ith\ element \\ 0 & elsewhere \end{cases} \quad \text{for i=1,2,3,.....r} \qquad (6)$$

and

$$\boldsymbol{H}(:,\boldsymbol{i}) = \boldsymbol{Yi} - \boldsymbol{Y0} \qquad for\ i = 1,2,3,\ldots\ldots\ldots.r \qquad (7)$$

Next by assuming, superposition law is satisfied, the channel matrix elements were identified from (7). Thereafter the channel matrix was resized into a standard pixel size of $M*N$ and stored. Standard resize channel matrix is denoted by $\boldsymbol{Hr}$

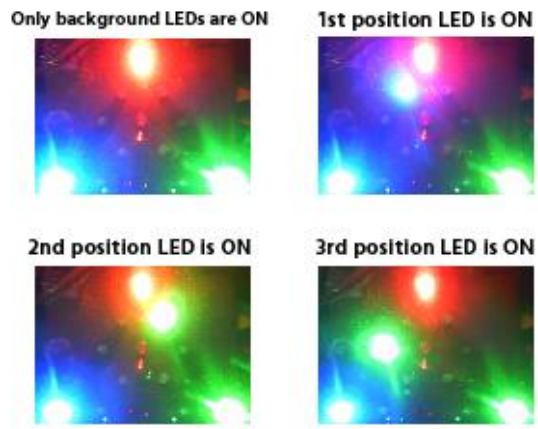

Fig. 8 Cropped images for channel identification in the experimental setup.

When transmission distance is increasing, the dimensions of the region of interest in the captured frame will reduce and vice versa. So effectively the number of pixels which are affected by the transmitter LEDs (number of output channels) decrease with increasing transmission distance. Channel matrix $\boldsymbol{H}$ will have different dimensions and different values for different transmission distances, and therefore we have,

$$\boldsymbol{H} = f(d) \quad (8)$$
$$size(\boldsymbol{H}) = f(d) \quad (9)$$

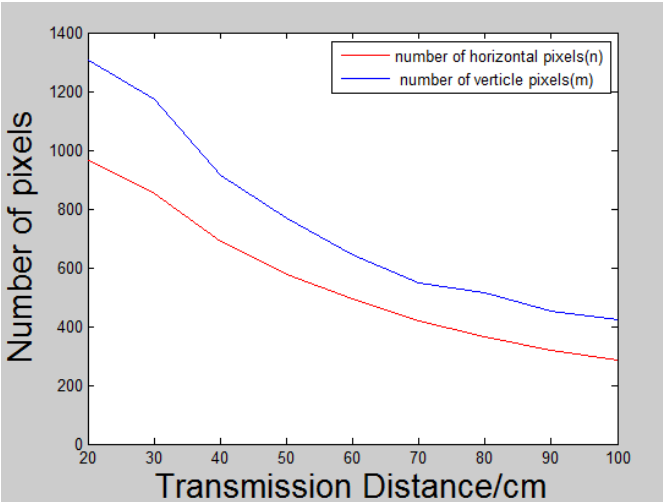

Fig. 9 Variation of no of vertical and horizontal pixel in the area of interest with varying distance

However through resizing channel matrixes corresponding to all distances are brought to the same size of *M*N*. But still resized channel matrixes have different values for different distances. Hence,

$$\boldsymbol{Hr} = f(d) \quad (10)$$

Thereafter for few transmission distances, ***Hr*** is identified and using some interpolation technique, resized channel matrix ***Hr*** was obtained as a function of the transmission distance.

General Algorithm flow for channel identification

for (at least 3 specific transmission distance)

- read (channel identification images)
- Crop the region of interest
- Resize the cropped region into M*N pixel region
- Generate a column vector from the pixels in the region
- Feed the known input
- Identify the corresponding channel matrix column values
- Identify the final channel matrix from equation (7)

end

#Generate the channel matrix as a function of distance using an interpolation method

#Store results

If the system is kept at the same noise level, water turbulent and same luminous level, the resized channel matrix function calculation in (10) can be used for data decoding purposes. To this end, we can use the following equation.

$$\boldsymbol{Xr} = inverse(transpose(\boldsymbol{Hr}(d)) * \boldsymbol{Hr}(d)) * transpose(\boldsymbol{Hr}(d)) * \boldsymbol{Yr} \quad (11)$$

where ***Yr*** is the given output, and ***Xr*** is the received input, ***Hr(d)*** is the resized channel matrix for the corresponding transmission distance calculated from resized channel matrix function. Transmission distance can be identified from the size of region of interest in the captured frame.

As *transpose* ***(Hr(d))*Hr(d)*** is singular, Moore-Penrose pseudo inverse is taken for quantity *transpose(**Hr(d))*Hr(d)**.* Finally some thresholding level is used to obtain a binary output vector as in (12)

$$Xrt = \begin{bmatrix} 1 \ if \ Xr \geq thr \\ 0 \ if \ Xr \leq thr \end{bmatrix} \quad (12)$$

where, ***Xrt*** is the calculated input values after thresholding, and *thr* is the threshold level.

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Algorithm flow for output Detection

- Read (image to be analyse)
- Crop the region of interest
- Identify the transmission distance from the cropped image size
- Resize the cropped region into M*N pixel region
- Obtain the resized channel matrix for that given transmission distance from resized system matrix function
- Generate a ***Yr*** vector from the pixels in resized image
- Obtain ***Xr*** from equation (11)
- Obtain the binary vector from equation (12)
- identify the input pattern

V. Evaluation of System Performance

A. *Bit Rate (BR)*

Bite rate is an important performance matrix that can be used to asses a communication system. Specifically bit rate is given by,

$$BR = r * f \quad (13)$$

where, r represent the number of transmission streams in the transmitter, and f is the frame rate of the camera.

B. Bit error rate (BER)

Errors in the system can occur if a frame is missed by the capture mechanism of the camera or due to erroneous identification of a LED pattern. If only rm number of LED states were incorrectly decoded, BER can be calculated from,

$$BER = \frac{rm}{s} * 100 \quad (14)$$

Moreover, if only fm number of frames are missed is a second, BER can be expressed as,

$$BER = \frac{fm}{f} * 100 \quad (15)$$

And if both fm number of frames are missed and rm number of LED states are decoded in error, BER can be calculated as,

$$BER = \frac{fm*r+(f-fm)*rm}{f*r} * 100 \quad (16)$$

We note that the BER is highly affected due to the auto focus feature of the used camera. In the next sub section we discuss this effect in detail.

C. Problem due to Auto Focus

The focus distance vary with the number of ON LEDs in the transmitter. This is due the automatic feature that the camera has in order to control the luminous level which it absorbs. Hence if auto focus option is used, for the same transmission distance the focus distance vary with number of LEDs present in the transmitter. This undesirable action of the camera, influences validity of the super position assumption that we considered in order to identify the system matrix in (7).

The effect of focus distance in the captured frame region of interests for a constant transmission distance for varying number of LEDs ON condition is shown in Fig. 10,

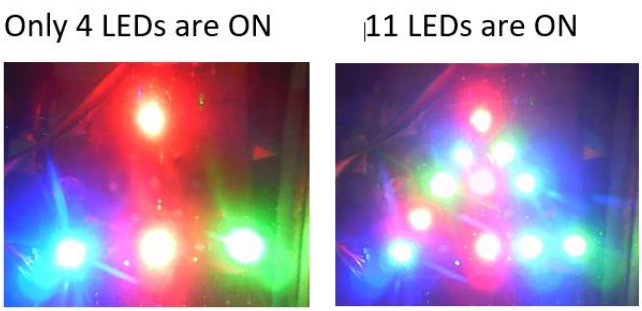

Fig. 10 The captured frames with different number of ON LEDs At a given distance

It is evident from Figure 10 that, when the number of ON LEDs increases, the spread of a given LED reduces. This problem was solved by, using manual focus setup in the camera. Focus distance was maintained at a fixed distance irrespective of transmission distance and number of ON LEDs in the transmitter. By doing so the spread of a given LED light was maintained at same level if it is ON, irrespective of the number of ON LEDs in the transmitter.

D. Camera Frame Rate and Transmitter Switching Frequency

Another effect that influences the BER of the VLC system is the transmitter pattern switching frequency. In a traditional single input single output VLC system, if Bit Rate of transmitter signal is BR, then at every $1/BR$ second, a data is decoded at the receiver. However, due to the presence of the camera in our design, if Bit Rate is BR, then at every $1/f$ second, r number of data bits are decoded. Here, $BR=f*r$ and f, r represent frame rate and the number of bit streams in the transmitter respectively. Therefore if one frame is missed it results in r number of data bits being missed, which significantly affect the bit rate. Hence it is recommended to use a camera which produces a video stream with constant frame rate.

VI. EXPERIMENTAL RESULTS

In this section we present BER results obtained using our experimental underwater VLC setup and an analysis of the accuracy of the resized channel matrix function calculated. Resized channel matrix function was obtained based on linear interpolation, using the system identification images corresponding to distances of 0.2m, 0.5m, and 1m. Thereafter in order to check the validity of the calculated resized channel matrix function, channel identification frames at different distance were fed in, and using (10) and (11), binary outputs were identified.

Table 1
Variation of erroneous frames detected and Frame error rate for different distances

Transmission Distance/cm	No of channel identification frames	No of erroneous frames detected	Frame Error Rate as a percentage(%)
20	12	0	0
30	12	0	0
40	12	0	0
50	12	0	0
60	12	2	16.67
70	12	1	8.33
80	12	0	0
90	12	1	8.33
100	12	0	0

Table 1 shows the resized channel matrix function having an accuracy of at least 83.33% for any given transmission distance.

Thereafter the resized system matrix function was used to decode inputs for random video stream which had considerable frames. The obtained results were tabulated in table 2.

Table 2
Variation of BER with distance of random frames

Transmission Distance/cm	No of Bites transmitted	No of error detected	Bit Error Rate(BER) as a %
20	300	27	15
30	300	22	13.33
40	300	16	9.33
50	300	18	8
60	300	22	10.33
70	300	25	12.33
80	300	27	15
90	300	30	19
100	300	43	22.33

From Table 2 it is evident that the BER increases with transmission distance. This is due to two reasons. They were, reduction of the pixels in the region of interest in the captured frames with increasing transmission distances and the due to the inability of the camera to focus into longer distances efficiently as compared to shorter distances.

The above BER performance were obtained when transmission data speeds were in the order of bits per second (bps). This low bit rate performance was due to the less number of LEDs at the transmitter. However, by increasing the number of LEDs and using a camera which has better resolution, higher data rates can be achieved.

The achieved BER performance is comparatively higher than the existing camera based VLC system BER performances [1]. Hence the proposed system could be used with high number of LEDs at the transmitter to realize a future high-speed underwater communication network which can yield better BER performance than the existing ones.

VII. CONCLUSION

A new under water VLC communication system design is presented. Transmitter is made of LEDs and a camera is used at the receiver. A practically implementable channel identification technique, output detection algorithm and system performance in terms of bit rate and bit error rate is presented. From the above study, it was realised that, even though BER increases with increasing transmission distances, reasonable BER performance can be achieved using the proposed system design compared to the existing alternatives for short and medium range underwater communications. Since this research is a preliminary study on underwater visible light communication based on a camera receiver, it is expected that the extension of this work on image processing for output detection, would give a better BER performance. Furthermore, the proposed system could be used with high number of LEDs at the transmitter to realize a future high-speed underwater communication network.

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